Private IP vs Public IP

What is a Private IP Address?

Definition

A **Private IP Address** is an IP address that is used **within a private network** (such as a home, office, or organization). It is **not directly accessible from the Internet**.

Private IP addresses are assigned to dPevices by a **router** or a **DHCP server**.

Why Do We Need a Private IP?

Private IP addresses allow devices within the same local network to communicate with each other without consuming public IP addresses.

Benefits

- Conserves Public IP addresses
- Improves network security
- Allows communication within a local network
- Free to use (No registration required)

Private IP Address Ranges

There are **three reserved ranges** for Private IP addresses.

Class	Private IP Range
Class A	**10.0.0.0 – 10.255.255.255**
Class B	**172.16.0.0 – 172.31.255.255**
Class C	**192.168.0.0 – 192.168.255.255**

Interview Tip: These three ranges are very important and are frequently asked in networking interviews.

Example

Suppose your laptop is connected to your home Wi-Fi.

```
Laptop
IP Address:
192.168.1.20
```

This is a **Private IP Address** because it belongs to the **192.168.x.x** range.

What is a Public IP Address?

Definition

A **Public IP Address** is a globally unique IP address assigned by an **Internet Service Provider (ISP)**. It is used to identify your network on the Internet and allows communication with devices across the world.

Unlike a Private IP, a Public IP is **accessible over the Internet**.

Why Do We Need a Public IP?

A Public IP allows your network to communicate with websites, cloud services, and servers on the Internet.

Without a Public IP, your network would not be able to access Internet resources.

Example

Your Internet Service Provider assigns your Wi-Fi router a Public IP such as:

```
49.205.120.45
```

When you visit **google.com**, your request is sent from this Public IP.

How Private and Public IP Work Together

Consider the following home network.

```
                Internet
                   |
                   |
         Public IP: 49.205.120.45
                   |
             Wi-Fi Router
          Private IP: 192.168.1.1
                   |
        ----------------------------
       |           |               |
Laptop          Mobile Phone       Smart TV
192.168.1.20    192.168.1.21     192.168.1.22
```

Explanation

- Each device inside the home network has its own **Private IP Address**.
- The Wi-Fi router has:
    - One **Private IP** (inside the network)
    - One **Public IP** (assigned by the ISP)
- When a device accesses the Internet, the router translates the Private IP into the Public IP using **NAT (Network Address Translation)**.

Note: NAT will be covered in detail in a later chapter.

Private IP vs Public IP

Feature	Private IP	Public IP
Scope	Local Network	Internet
Accessible from Internet	❌ No	✅ Yes
Assigned By	Router / DHCP Server	Internet Service Provider (ISP)

Feature	Private IP	Public IP
Unique	Unique within the local network	Globally Unique
Cost	Free	Provided by ISP
Example	192.168.1.20	49.205.120.45

Real-World Example

Suppose you open **youtube.com**.

```
Laptop
192.168.1.20
      │
      ▼
Wi-Fi Router
Public IP
49.205.120.45
      │
      ▼
Internet
      │
      ▼
YouTube Server
```

Your laptop sends the request using its **Private IP**, but the router forwards it to the Internet using its **Public IP**.

Quick Revision

Private IP	Public IP
Used inside a local network	Used on the Internet
Not accessible from the Internet	Accessible from anywhere on the Internet
Assigned by Router/DHCP	Assigned by ISP
Three Reserved Ranges	Globally Unique

Interview Corner

Q1. What is a Private IP Address?

Answer:

A Private IP Address is an IP address used within a private network. It is not directly accessible from the Internet.

Q2. What is a Public IP Address?

Answer:

A Public IP Address is a globally unique IP address assigned by an Internet Service Provider (ISP) that enables communication over the Internet.

Q3. Who assigns a Private IP Address?

Answer:

A Router or DHCP Server.

Q4. Who assigns a Public IP Address?

Answer:

An Internet Service Provider (ISP).

Q5. Can a Private IP Address be accessed directly from the Internet?

Answer:

No.

Q6. Name the three Private IP ranges.

Answer:

- **10.0.0.0 – 10.255.255.255**
- **172.16.0.0 – 172.31.255.255**
- **192.168.0.0 – 192.168.255.255**

Q7. Which technology converts a Private IP into a Public IP?

Answer:

NAT (Network Address Translation).

Key Takeaways

- A **Private IP Address** is used inside a local network and cannot be accessed directly from the Internet.
- A **Public IP Address** is assigned by an ISP and is used for Internet communication.
- Every device in your home network has a **Private IP**, while the router has a **Public IP**.
- **NAT** enables devices with Private IP addresses to communicate with the Internet using a Public IP.
- Remember the **three Private IP ranges**, as they are commonly asked in interviews.